

3.6 Sustainability Systems Add-Alternate — 1212 Lobdel Avenue

All three systems designed for fee-simple individual lot ownership. Costs indexed to 54-unit Option D yield. PACE financing attaches to individual properties. HOA covers shared infrastructure under recorded easement plat.

System costs & primary benefit

SYSTEM	TOTAL COST	PER UNIT	PRIMARY BENEFIT
Geothermal HVAC <i>Shared bore field · 37 vertical bores Drilled under drive aisles</i>	\$672,217 <i>vs \$459,000 conventional</i>	\$12,448	4.4-yr simple payback · \$900/unit/yr energy savings · 3.5 COP vs 1.0 gas
Community Solar PV <i>191 kW aisle canopy 267,120 kWh/yr · 66% community energy</i>	\$599,040 gross \$419,328 net (30% ITC)	\$7,765 net	Virtual net metering (Delmarva) · \$693/unit/yr savings · Covered parking
Rainwater Harvesting <i>6,000-gal underground cistern Irrigation only · Under center aisle</i>	\$26,820	\$497	1,014,292 gal/yr collection · Stormwater credit · Enterprise Green Communities pts
Combined total	\$1,118,365	\$20,710/unit	All three systems · PACE-eligible · Day-one cash flow positive

HOA shared maintenance budget

LINE ITEM	ANNUAL	\$/UNIT/MO
Geothermal loop maintenance	\$2,400	\$3.70
Solar PV (inverters, cleaning)	\$3,200	\$4.94
Rainwater system	\$1,200	\$1.85
Drive aisle maintenance	\$4,000	\$6.17
Landscaping + irrigation	\$8,500	\$13.12
Insurance	\$3,600	\$5.56
Reserve fund + management	\$4,690	\$7.24
Total HOA	\$27,590	\$43/mo

Monthly cash flow per unit

MONTHLY COSTS	
PACE — geothermal premium	-\$28
PACE — solar (net of ITC)	-\$54
HOA fee (all systems)	-\$43
Total out	\$125/mo
MONTHLY SAVINGS	
Energy savings — geothermal	+\$75
Solar net metering credits	+\$58
Water savings — irrigation	+\$1
Total savings	\$134/mo

Net per unit per month — savings minus all costs

PACE payments + HOA fee fully covered by energy and solar savings on day one

+\$9 / mo

Monthly flow by component (54-unit Option D)

PACE — geothermal		-\$28
PACE — solar (net ITC)		-\$54
HOA fee		-\$43
Geo energy savings		+\$75
Solar net metering		+\$58
Water savings		+\$1
Net		+\$9

FHA mortgage qualification check (\$215K purchase · 6.5% · 30 yr)

Principal + interest \$1,359 per month	HOA monthly \$43 per month	Taxes + insurance (est.) \$250 per month
Total PITI + HOA \$1,652 per month	Income required (FHA 31%) \$63,930 per year	80% AMI (family of 3) ✓ Qualifies ~\$68,000/yr at 80% AMI

Sustainability costs scaled to each yield option

YIELD OPTION	UNITS	GEO TOTAL	SOLAR (NET ITC)	RAINWATER	COMBINED	PER UNIT	HOA/MO
Option B (full indep. setbacks)	36	\$448,145	\$419,328	\$26,820	\$894,293	\$24,841	\$45
Opt A+FW12 (cross street + fire wall)	33	\$411,466	\$419,328	\$26,820	\$857,614	\$25,988	\$47
Option D (shared easements)	54	\$672,217	\$419,328	\$26,820	\$1,118,365	\$20,710	\$43

Solar canopy and rainwater system are largely fixed to site infrastructure; geo scales with units. Option D benefits most from scale — lowest per-unit cost and HOA.

⚠ Confirm with Energize Delaware that PACE financing attaches to individual fee-simple lots, not PUD/condo structure only.

⚠ Confirm Delmarva Power virtual net metering participation for HOA subscriber structure before committing solar to aisle canopy configuration.

⚠ HOA covenant documents must cover geothermal loop access rights (bore field under drive aisles), solar canopy maintenance responsibilities, and rainwater system. Attorney review required before plat recording.